

Welcome to 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

Grades & Deadlines

- To receive your A1 programming grade: make sure to fill out the PL question
- Please submit your project proposal updates if you haven't already. If you need more time, please make a private Piazza post to let me know when this week you are planning to submit it. You will still be expected to meet your goal for the Week 4 checkpoint you set up for yourself.
- A2 Threshold for CNN baseline: updated to 0.74 (may be further updated to be lower, but no more than 0.74)

Symbolic AI

- Knowledge is written down, not learned. Facts and rules.
- Thinking is manipulating symbols.
- You can read the reasoning. Every step is inspectable, explainable
- Great when the rules are known: chess, tax law, logistics, theorem proving
- Breaks on messier/harder to define tasks: vision, language, anything with exceptions
- Remember we talked about epistemology—symbolic AI is good *when you know how you know*.

**IF fever AND stiff_neck
THEN suspect_meningitis**

Connectionist AI

- Knowledge is learned, not written
- Learned from examples, by adjusting connections (weights)
- Thinking is patterns lighting up. Many simple units, not centralized
- Nobody can read it — the knowledge is smeared (distributed) across millions of numbers
- Great when the rules are unknown: vision, speech, language, faces
- Breaks in ways you can't see coming: confident, wrong, unfixable by hand
- Connectionist AI is useful when *you don't know how you know*.

**How do you recognize a friend's face?
You probably don't know.**

How do you know if your model is doing a task well?

Let's say you want to model cognition. How would you know if your model is good?

- A model that has the same output as humans isn't necessarily doing what humans are doing.
 - Two different approaches could be right—which approach is your brain doing?
 - It has to be wrong the way people are wrong.
- Errors as evidence: kids say "goed" instead of "went"; the network says "goed" too, unprompted
- Speed as evidence: you read letters faster inside words than alone; the model predicts by how much
- Development as evidence: the model passes through the same stages, in the same order
- Damage as evidence: lesion the network, get patterns matching real brain injury
- A metric: can your model predict the failures you didn't design for?

Guest lecture from Jay McClelland on Monday–be on time!

THE PARALLEL DISTRIBUTED PROCESSING APPROACH TO SEMANTIC COGNITION

James L. McClelland^{} and Timothy T. Rogers[‡]*

How do we know what properties something has, and which of its properties should be generalized to other objects? How is the knowledge underlying these abilities acquired, and how is it affected by brain disorders? Our approach to these issues is based on the idea that cognitive processes arise from the interactions of neurons through synaptic connections. The knowledge in such interactive and distributed processing systems is stored in the strengths of the connections and is acquired gradually through experience. Degradation of semantic knowledge occurs through degradation of the patterns of neural activity that probe the knowledge stored in the connections. Simulation models based on these ideas capture semantic cognitive processes and their development and disintegration, encompassing domain-specific patterns of generalization in young children, and the restructuring of conceptual knowledge as a function of experience.

For Monday's Prelecture Homework: Post on Piazza

Discussion Prompt for July 20: Semantic Cognition (Due Jul 19)

Updated 41 seconds ago by Annapurna Vadaparty

In class on Mon, July 20, we will be discussing a cognitive model that relies on back propagation: a model of Semantic Cognition introduced by Rumelhart, and then later applied to modeling development of semantic cognitive abilities and their disintegration in dementia by Tim Rogers and Jay McClelland.

To prepare for class, read the [paper by McClelland and Rogers on the Parallel Distributed Processing Approach to Semantic Cognition](#). Prepare a prep statement by Jul 19 (1-2 paragraphs, up to 400 words). **Post your response here as a follow-up.**

In your prep statement, you could write about one of the following, or raise other any other thoughts or questions you may have:

1. Discuss the problems with the hierarchical propositional model that motivated the PDP approach and how it addresses these problems.
2. Consider one feature of semantic cognition that the model addresses and discuss the pros and cons of the model's approach.
3. Consider the model as a computational engine for semantic cognition. What strengths and weaknesses does it have, and for the weaknesses, how might you address them?

Prof. McClelland will go over some of the motivations and results in class, and our discussion will be guided by your responses here.

Transformer Networks (NeurIPS 2017)

Attention Is All You Need

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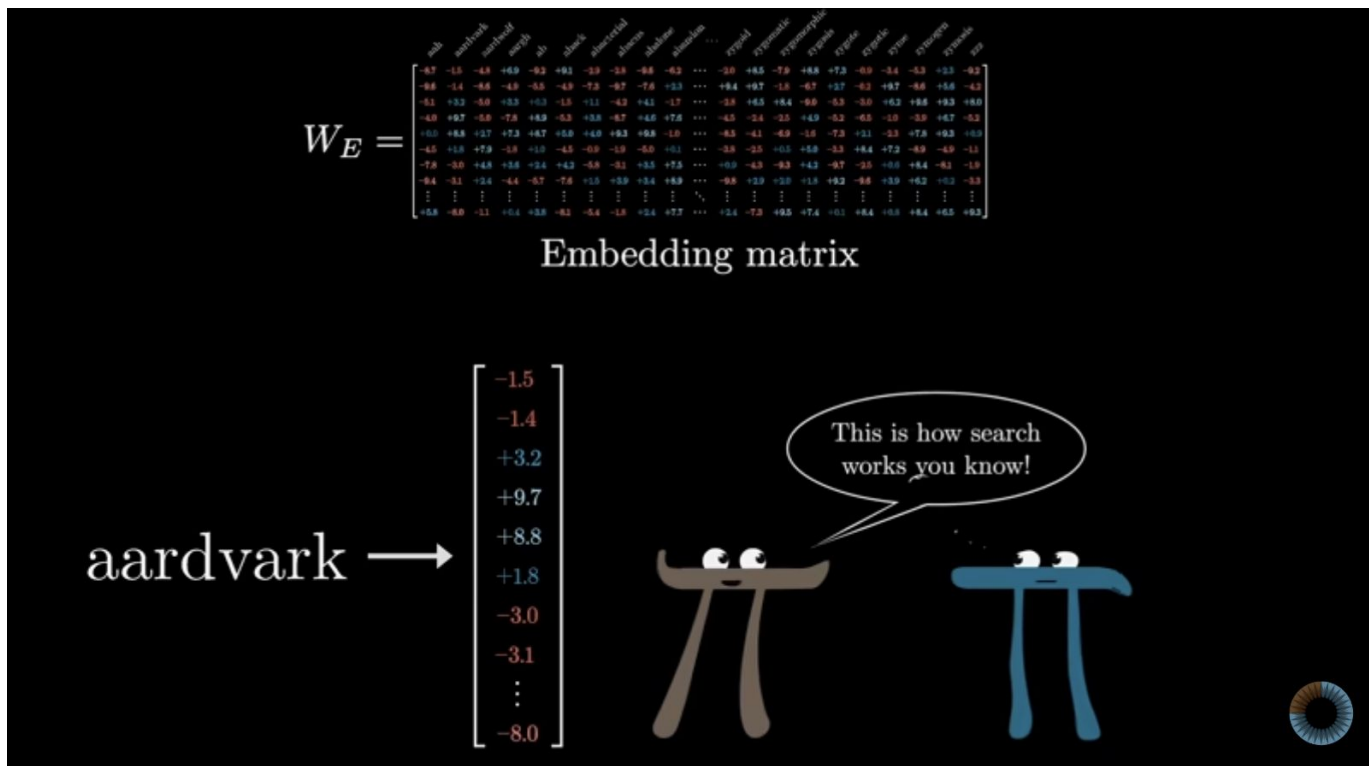
Who has heard of this paper?

Who knew that it was a reference to the Beatles song “Love is All You Need”?

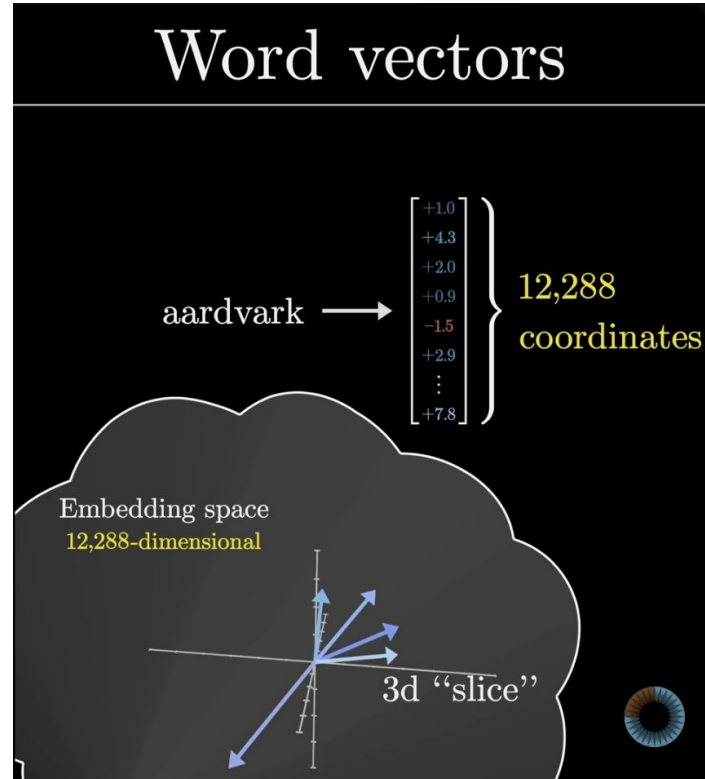
Now that we have these properties of language...

1. What is our data and why? **Why language** ✓
2. How are we representing the data? **Tokens** ✓ and **embeddings**
3. What is the architecture of the model we're using? **Transformers**
4. How does this architecture work? **Attention**

Embeddings (have been around before transformers)



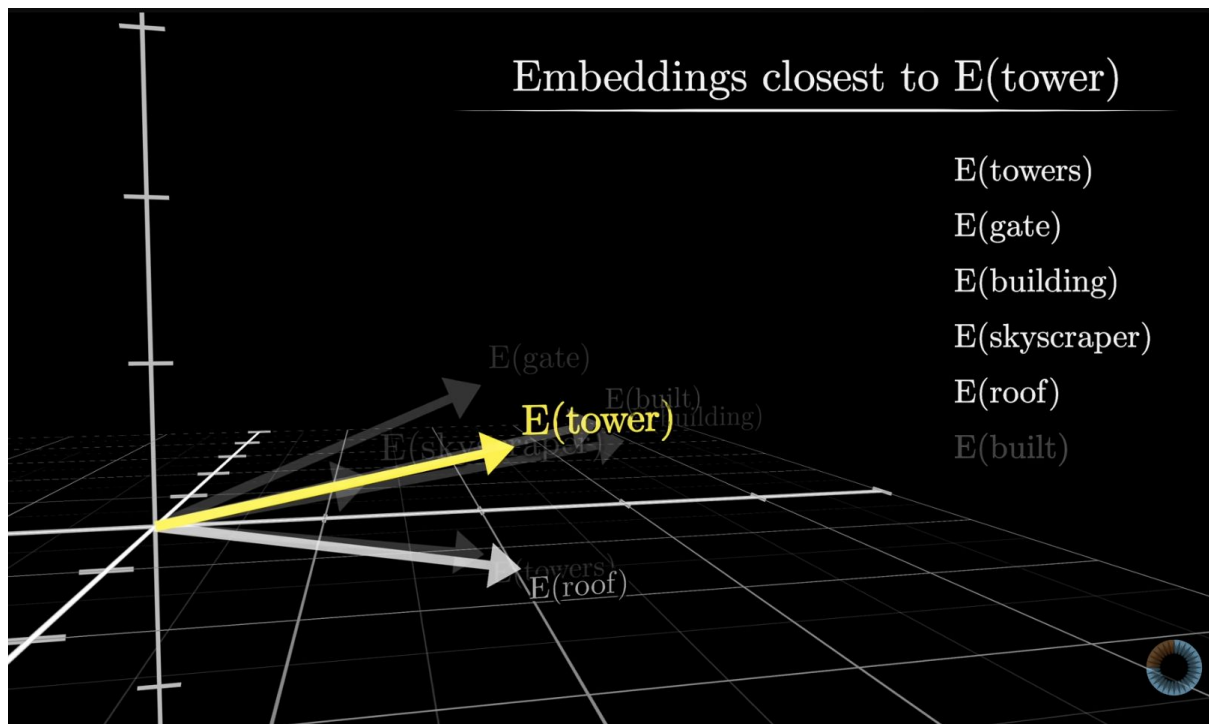
A note about visualizations



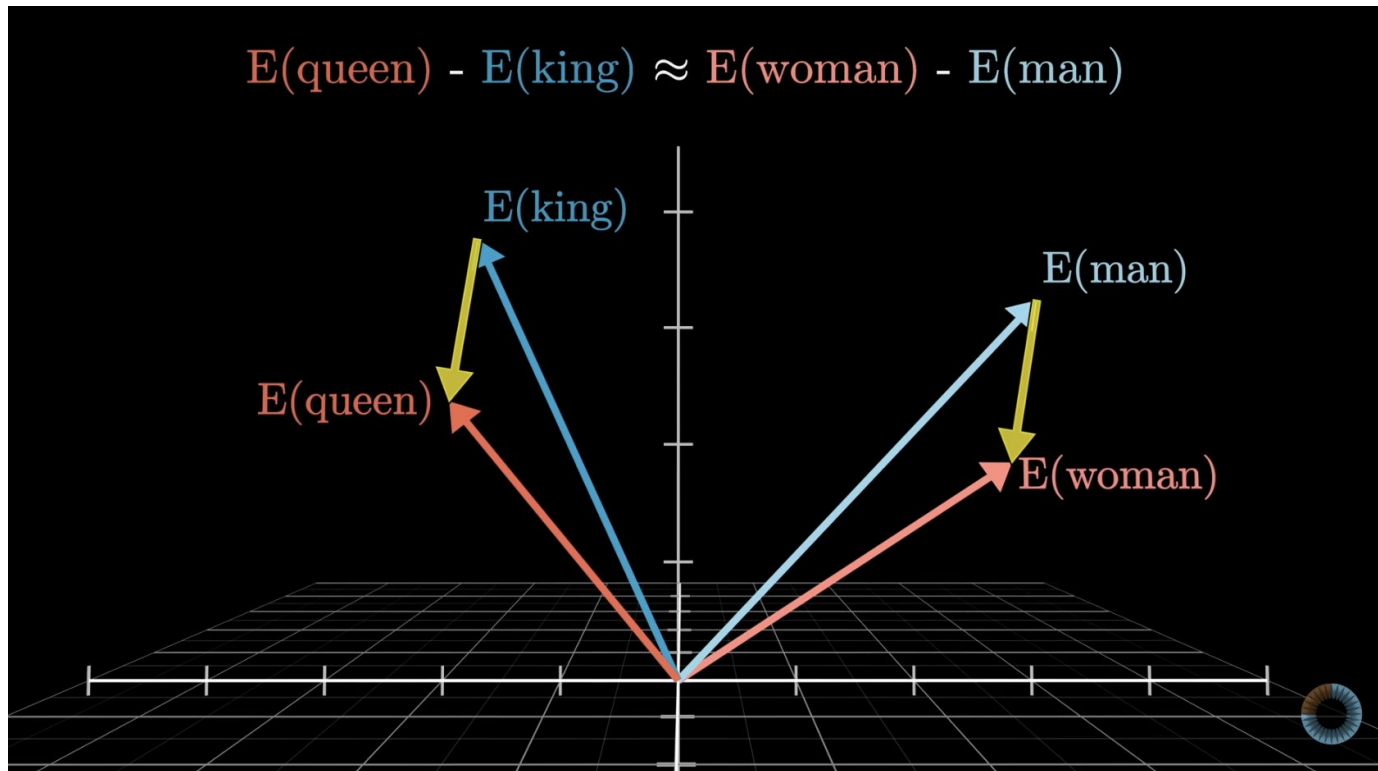
Why do we want such a high dimensional embedding?

- The size of the embedding vector (e.g. 12,288) is how many unique attributes of the embedding we can represent
- Maybe for an object, things like color, size, sentiment, where you encounter it...these are just examples, but having more gives you more expressive representations

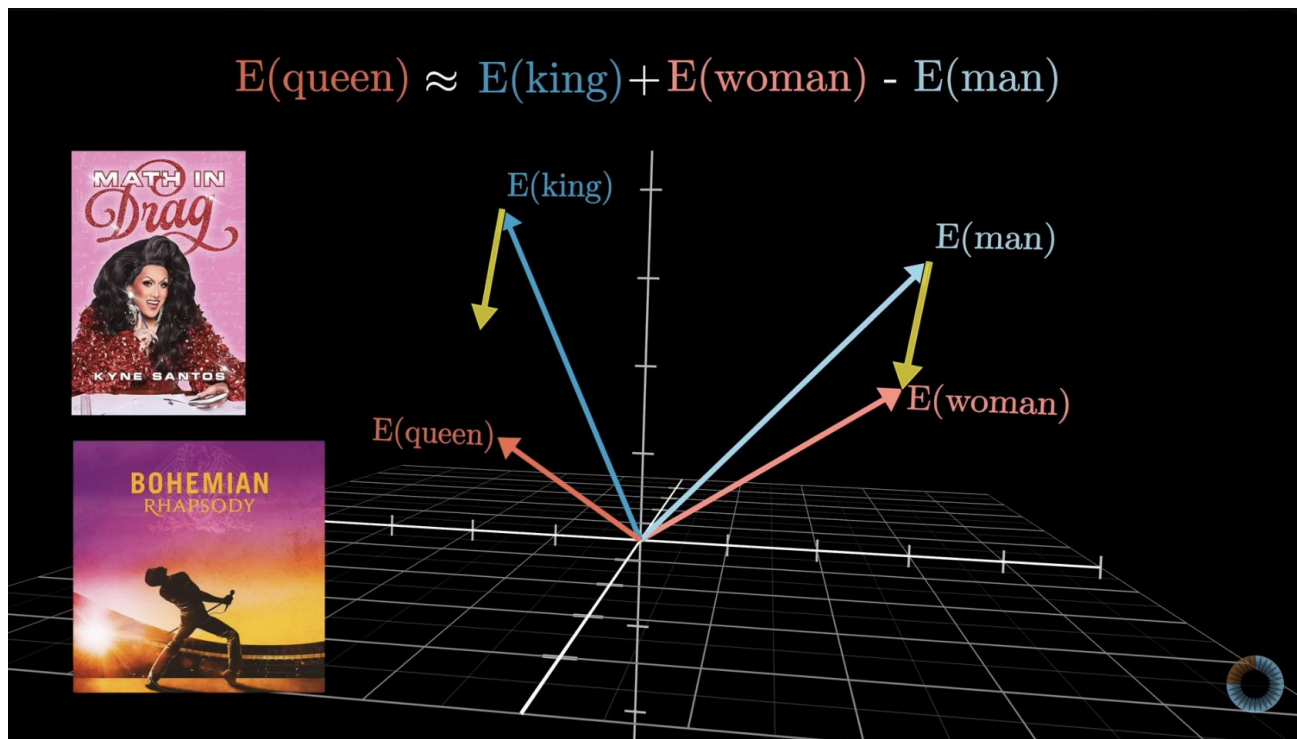
Embeddings & Meaning



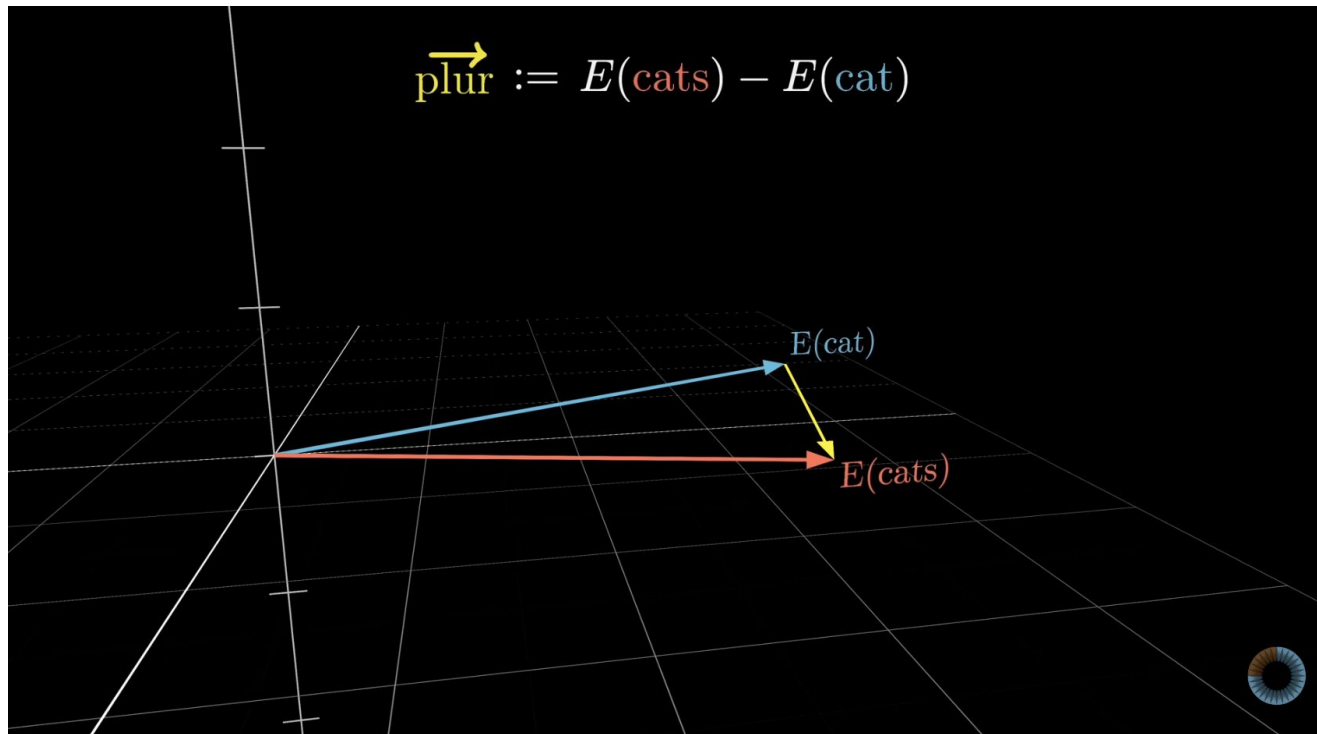
What are these embeddings telling us?



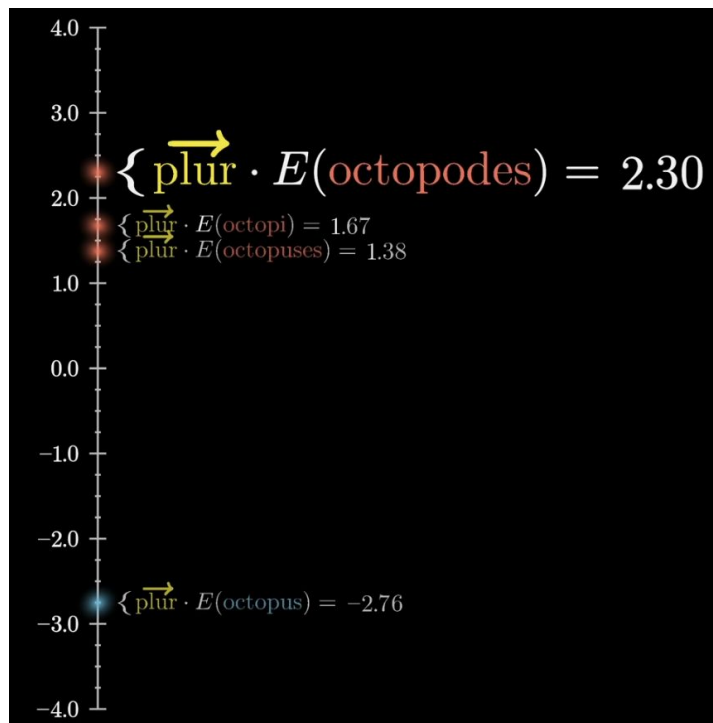
Not exactly for overloaded words like “queen”



Example: Hypothesizing a plurality vector



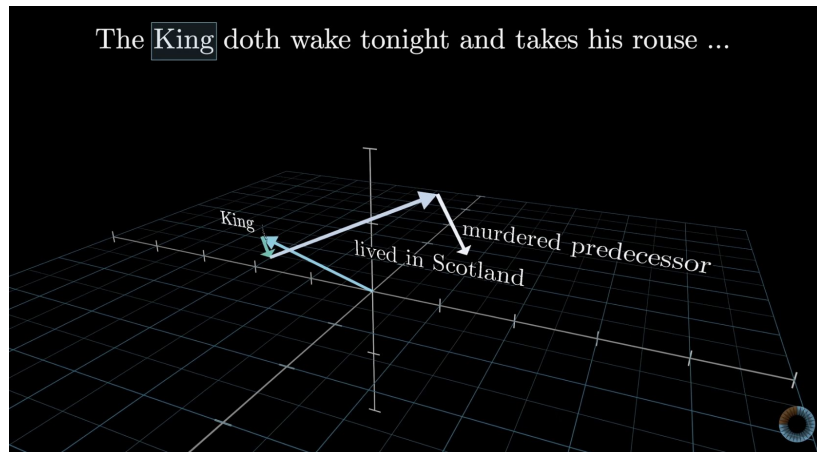
How do we test this? What do we find?



What are embeddings for?

*“You should think of [embeddings] as having the **capacity to soak in context**. A vector that started its life as the embedding of the word ‘king’, for example, might **progressively get tugged and pulled** by various blocks in this network, so that by the end, it points in a **much more specific and nuanced direction that somehow encodes that it was a king who lives in Scotland...**”*

(This again reminds us why we want lots of available dimensions...more nuance available.)

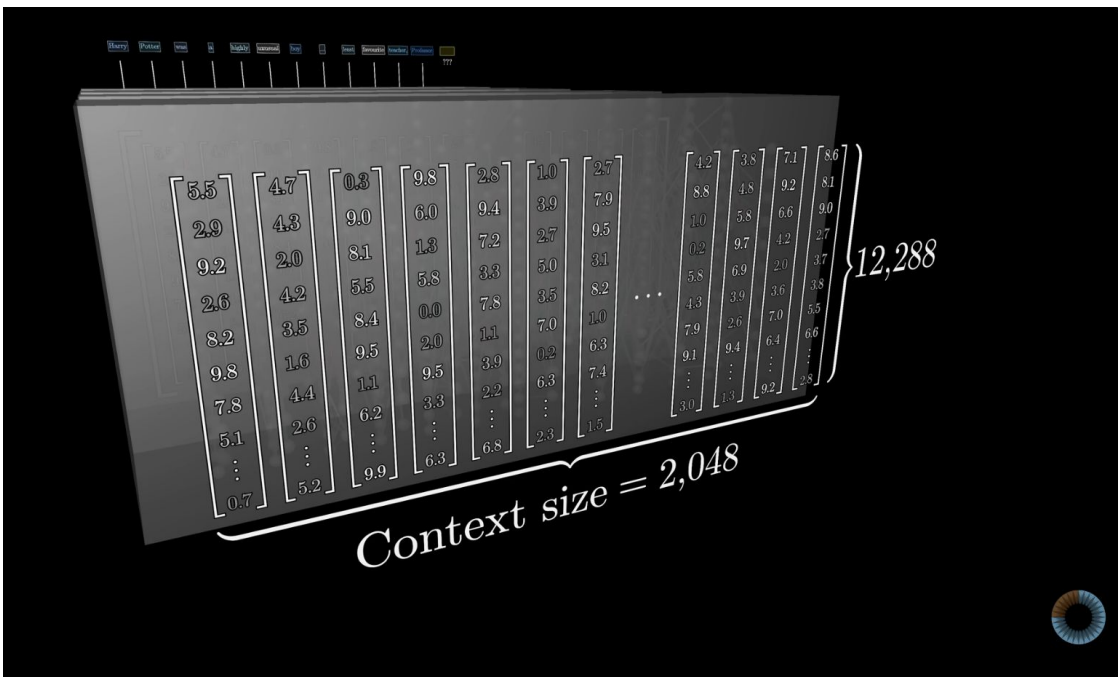


Now that we have these properties of language...

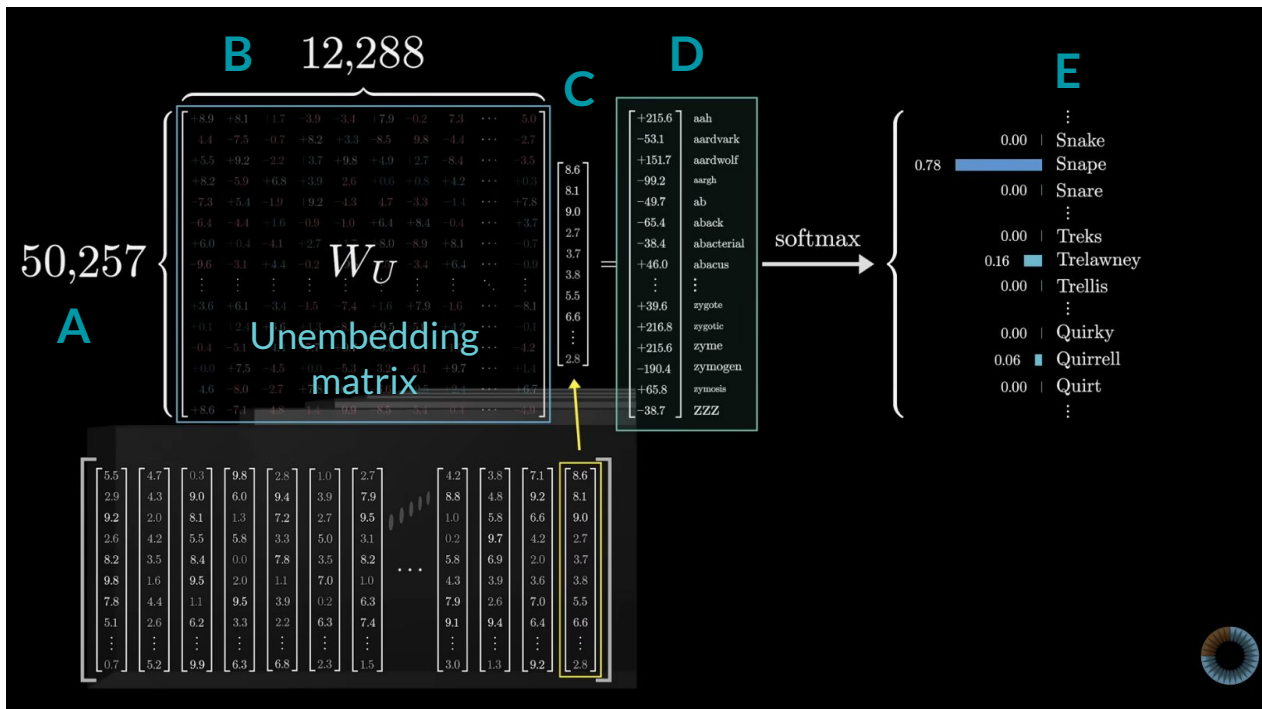
1. What is our data and why? **Why language** ✓
2. How are we representing the data? **Tokens** ✓ and **embeddings** ✓
3. What is the architecture of the model we're using? **Transformers**
4. How does this architecture work? **Attention**

What is “context”?

- From yesterday—what did we say about the “mapping inputs to space” strategy? What do we have to know about the inputs?



What do each of these dimensions/vectors (A-E) represent?
Chat with your group.



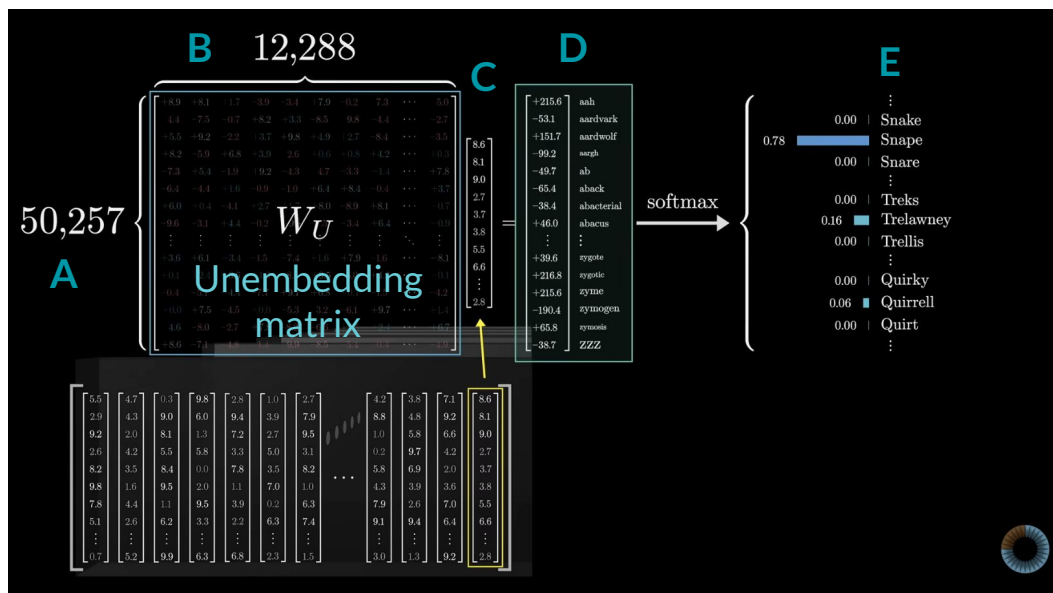
What do each of these dimensions/vectors represent?

B:
“Unembedding”
(one row for each token)

C:
Embedding of last
token in the context

D:
A number (pre-probability/logit)
for each token in the vocabulary

A:
tokens in the
vocabulary

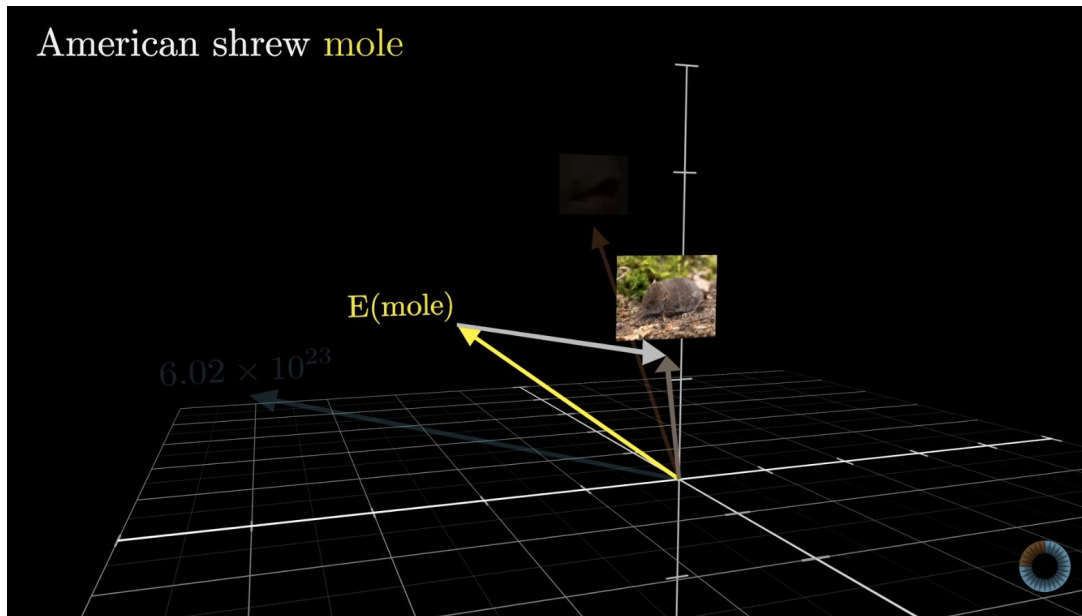


Now that we have these properties of language...

1. What is our data and why? **Why language** ✓
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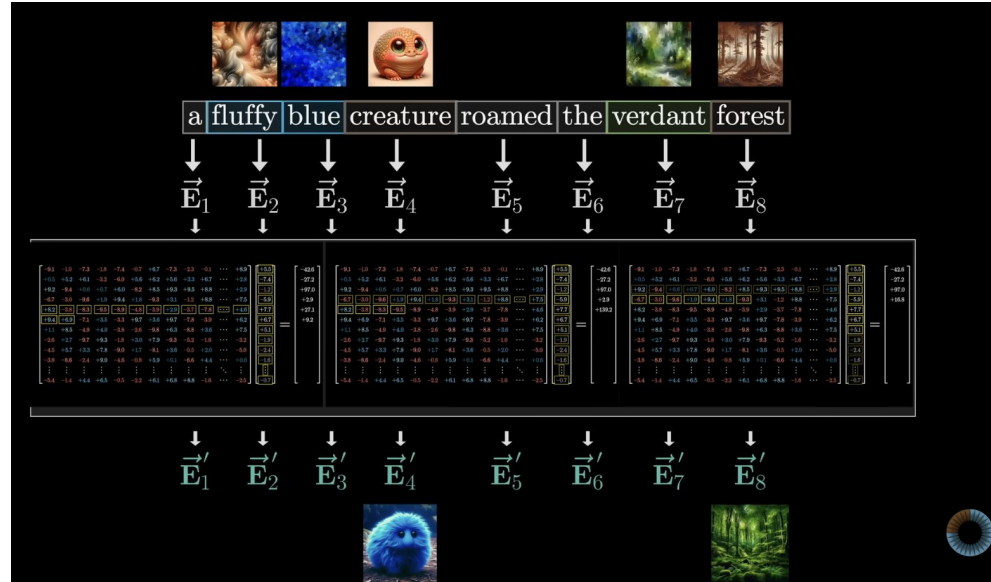
What is the goal of attention?

- We want each token's starting embedding to soak up all the context
- This means we want to figure out how much to move the initial embedding into the correct, context-rich embedding



How can we achieve this?

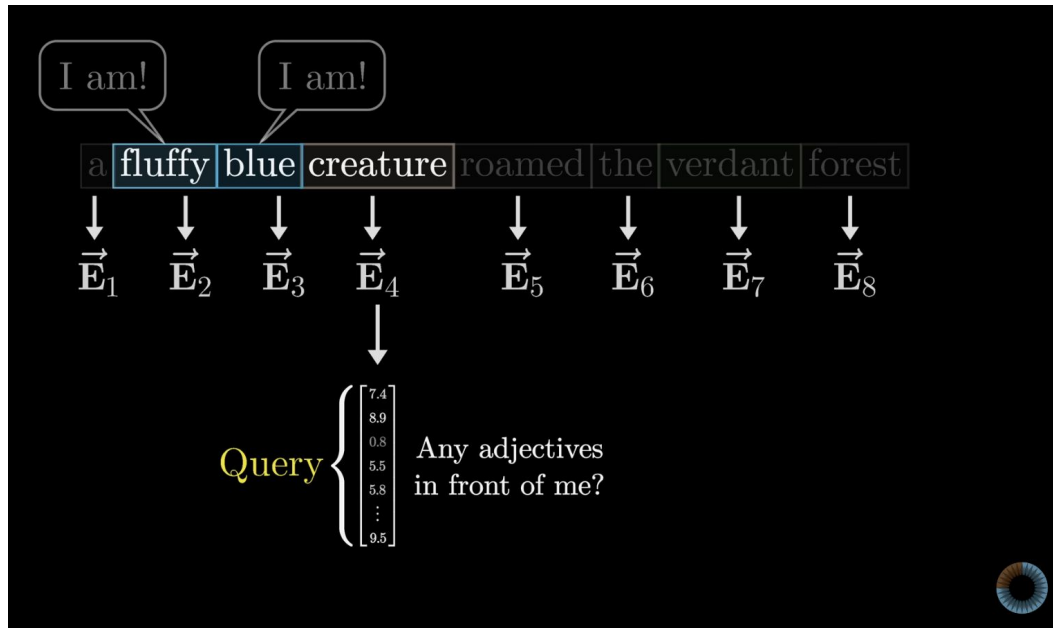
Let's see how we can transform one embedding into another using—as always—matrix multiplication.



3b1b, "Attention in transformers, step-by-step"

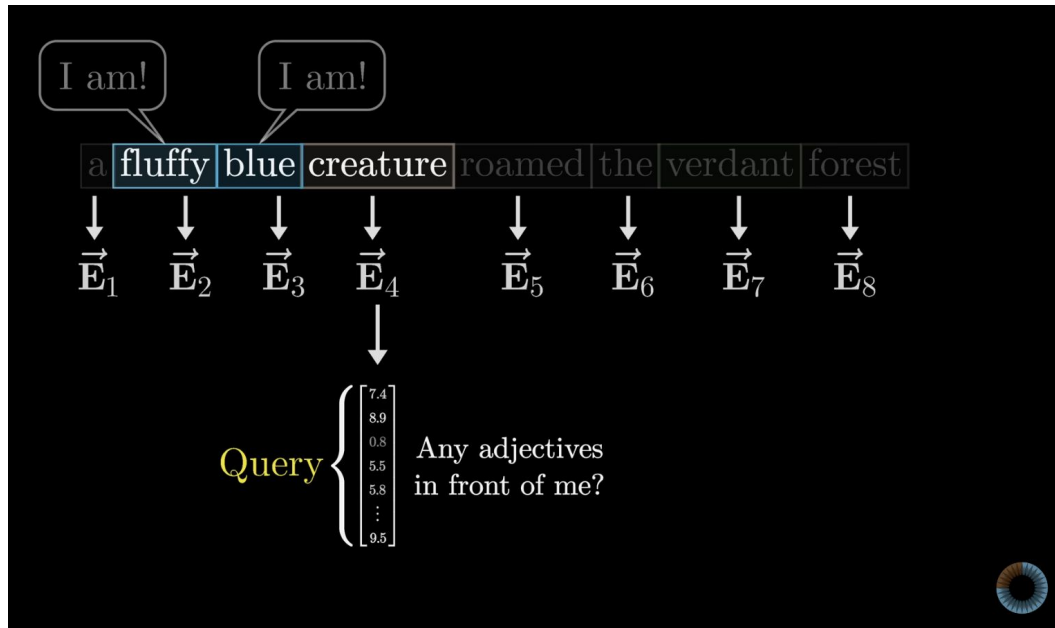
One useful vector: the Query (Q)

- What is the query doing?
- What part of this is depiction is accurate/not accurate?
- How might you phrase it instead?



One useful vector: the Query (Q)

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Creating Q

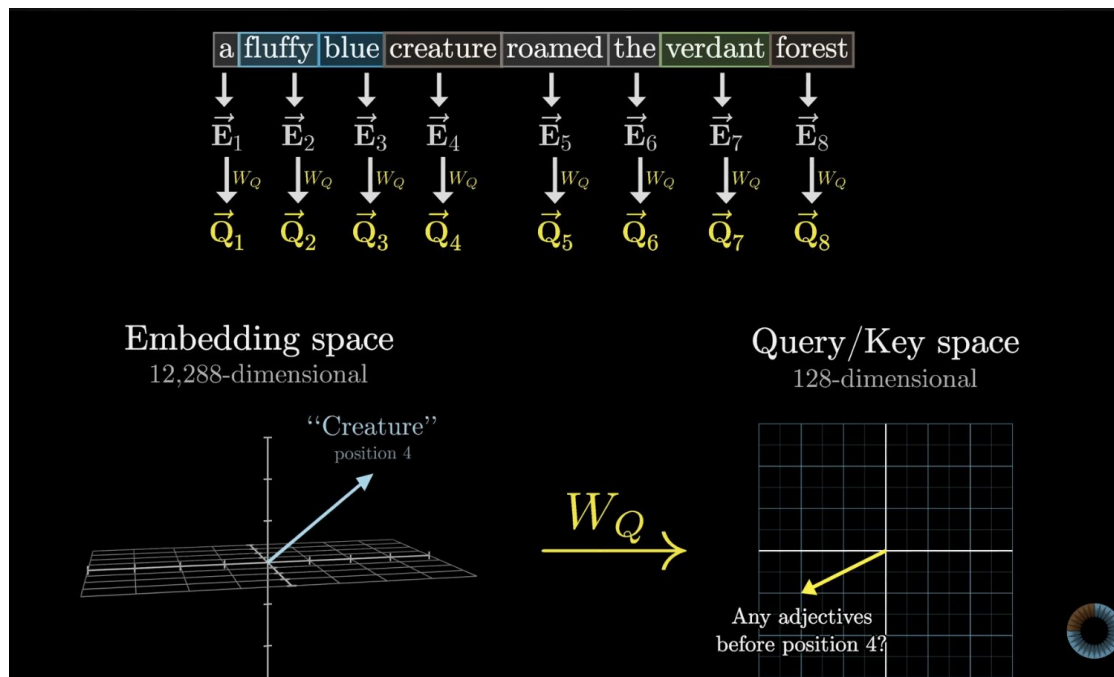
“The entries of this matrix are parameters of the model which means that the true behavior is learned from data, and in practice, what this matrix does at a particular attention head is challenging to parse.”

- Kind of like how PCA doesn't necessarily map onto components that “mean” something to us
- Explainable AI is interesting!

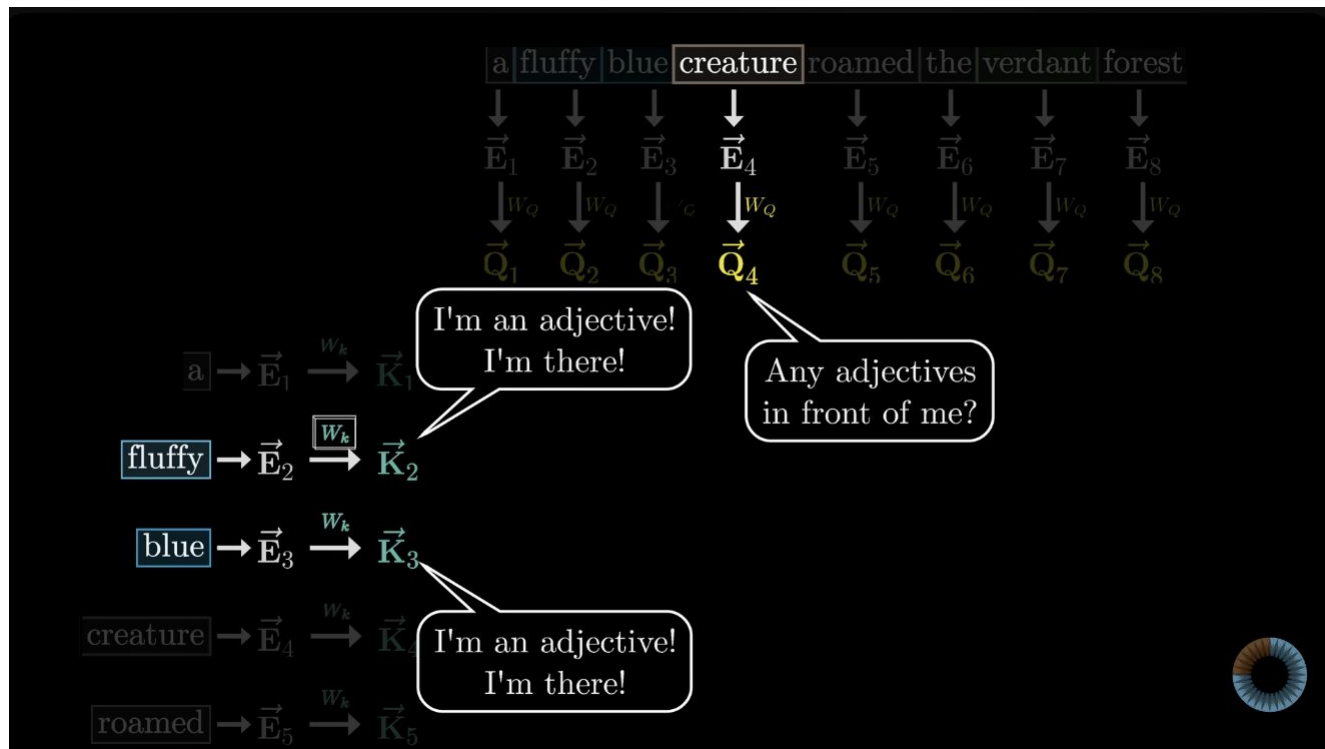
$$\overrightarrow{E}_i W_Q = \overrightarrow{Q}_i$$

$\overrightarrow{E}_i$	$\overrightarrow{Q}_i$
2.9	+310.6
2.4	-95.2
1.0	-2.1
0.2	-152.0
9.2	-123.2
6.6	⋮
7.8	-12.7
2.8	⋮
5.8	⋮
0.6	⋮
⋮	⋮
9.7	⋮

(An imagined) Query/Key space



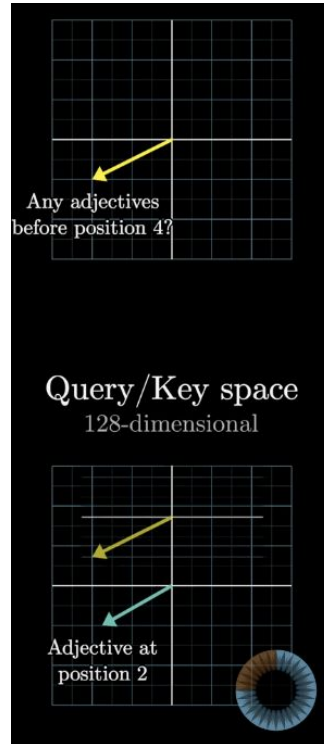
Keys: conceptually, “answering” queries



- How would we “answer” a vector?
- Hint: what do we do to check if vectors “agree”?



Query and Key matching: closely aligned vectors



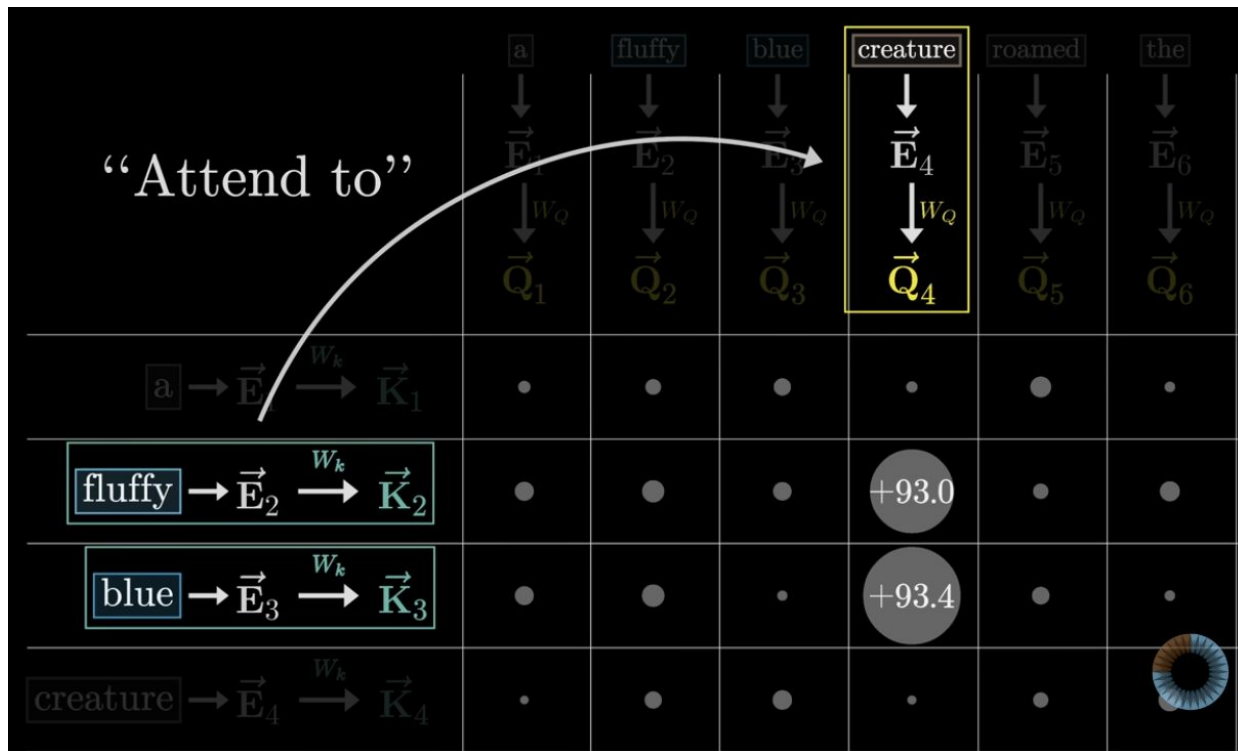
Now we do it for every key/query pair in the context

	a	fluffy	blue	creature	roamed	the	verdant	forest	
	$\downarrow$ $\vec{E}_1$ $\downarrow_{W_Q}$ $\vec{Q}_1$	$\downarrow$ $\vec{E}_2$ $\downarrow_{W_Q}$ $\vec{Q}_2$	$\downarrow$ $\vec{E}_3$ $\downarrow_{W_Q}$ $\vec{Q}_3$	$\downarrow$ $\vec{E}_4$ $\downarrow_{W_Q}$ $\vec{Q}_4$	$\downarrow$ $\vec{E}_5$ $\downarrow_{W_Q}$ $\vec{Q}_5$	$\downarrow$ $\vec{E}_6$ $\downarrow_{W_Q}$ $\vec{Q}_6$	$\downarrow$ $\vec{E}_7$ $\downarrow_{W_Q}$ $\vec{Q}_7$	$\downarrow$ $\vec{E}_8$ $\downarrow_{W_Q}$ $\vec{Q}_8$	
$\boxed{\text{a}} \rightarrow \vec{E}_1 \xrightarrow{W_K} \vec{K}_1$	$\vec{K}_1 \cdot \vec{Q}_1$	$\vec{K}_1 \cdot \vec{Q}_2$	$\vec{K}_1 \cdot \vec{Q}_3$	$\vec{K}_1 \cdot \vec{Q}_4$	$\vec{K}_1 \cdot \vec{Q}_5$	$\vec{K}_1 \cdot \vec{Q}_6$	$\vec{K}_1 \cdot \vec{Q}_7$	$\vec{K}_1 \cdot \vec{Q}_8$	
$\boxed{\text{fluffy}} \rightarrow \vec{E}_2 \xrightarrow{W_K} \vec{K}_2$	$\vec{K}_2 \cdot \vec{Q}_1$	$\vec{K}_2 \cdot \vec{Q}_2$	$\vec{K}_2 \cdot \vec{Q}_3$	$\vec{K}_2 \cdot \vec{Q}_4$	$\vec{K}_2 \cdot \vec{Q}_5$	$\vec{K}_2 \cdot \vec{Q}_6$	$\vec{K}_2 \cdot \vec{Q}_7$	$\vec{K}_2 \cdot \vec{Q}_8$	
$\boxed{\text{blue}} \rightarrow \vec{E}_3 \xrightarrow{W_K} \vec{K}_3$	$\vec{K}_3 \cdot \vec{Q}_1$	$\vec{K}_3 \cdot \vec{Q}_2$	$\vec{K}_3 \cdot \vec{Q}_3$	$\vec{K}_3 \cdot \vec{Q}_4$	$\vec{K}_3 \cdot \vec{Q}_5$	$\vec{K}_3 \cdot \vec{Q}_6$	$\vec{K}_3 \cdot \vec{Q}_7$	$\vec{K}_3 \cdot \vec{Q}_8$	
$\boxed{\text{creature}} \rightarrow \vec{E}_4 \xrightarrow{W_K} \vec{K}_4$	$\vec{K}_4 \cdot \vec{Q}_1$	$\vec{K}_4 \cdot \vec{Q}_2$	$\vec{K}_4 \cdot \vec{Q}_3$	$\vec{K}_4 \cdot \vec{Q}_4$	$\vec{K}_4 \cdot \vec{Q}_5$	$\vec{K}_4 \cdot \vec{Q}_6$	$\vec{K}_4 \cdot \vec{Q}_7$	$\vec{K}_4 \cdot \vec{Q}_8$	
$\boxed{\text{roamed}} \rightarrow \vec{E}_5 \xrightarrow{W_K} \vec{K}_5$	$\vec{K}_5 \cdot \vec{Q}_1$	$\vec{K}_5 \cdot \vec{Q}_2$	$\vec{K}_5 \cdot \vec{Q}_3$	$\vec{K}_5 \cdot \vec{Q}_4$	$\vec{K}_5 \cdot \vec{Q}_5$	$\vec{K}_5 \cdot \vec{Q}_6$	$\vec{K}_5 \cdot \vec{Q}_7$	$\vec{K}_5 \cdot \vec{Q}_8$	
$\boxed{\text{the}} \rightarrow \vec{E}_6 \xrightarrow{W_K} \vec{K}_6$	$\vec{K}_6 \cdot \vec{Q}_1$	$\vec{K}_6 \cdot \vec{Q}_2$	$\vec{K}_6 \cdot \vec{Q}_3$	$\vec{K}_6 \cdot \vec{Q}_4$	$\vec{K}_6 \cdot \vec{Q}_5$	$\vec{K}_6 \cdot \vec{Q}_6$	$\vec{K}_6 \cdot \vec{Q}_7$	$\vec{K}_6 \cdot \vec{Q}_8$	
$\boxed{\text{verdant}} \rightarrow \vec{E}_7 \xrightarrow{W_K} \vec{K}_7$	$\vec{K}_7 \cdot \vec{Q}_1$	$\vec{K}_7 \cdot \vec{Q}_2$	$\vec{K}_7 \cdot \vec{Q}_3$	$\vec{K}_7 \cdot \vec{Q}_4$	$\vec{K}_7 \cdot \vec{Q}_5$	$\vec{K}_7 \cdot \vec{Q}_6$	$\vec{K}_7 \cdot \vec{Q}_7$	$\vec{K}_7 \cdot \vec{Q}_8$	
$\boxed{\text{forest}} \rightarrow \vec{E}_8 \xrightarrow{W_K} \vec{K}_8$	$\vec{K}_8 \cdot \vec{Q}_1$	$\vec{K}_8 \cdot \vec{Q}_2$	$\vec{K}_8 \cdot \vec{Q}_3$	$\vec{K}_8 \cdot \vec{Q}_4$	$\vec{K}_8 \cdot \vec{Q}_5$	$\vec{K}_8 \cdot \vec{Q}_6$	$\vec{K}_8 \cdot \vec{Q}_7$	$\vec{K}_8 \cdot \vec{Q}_8$	

- What do large dot products mean?



Large dot product: this key “attends to” this query



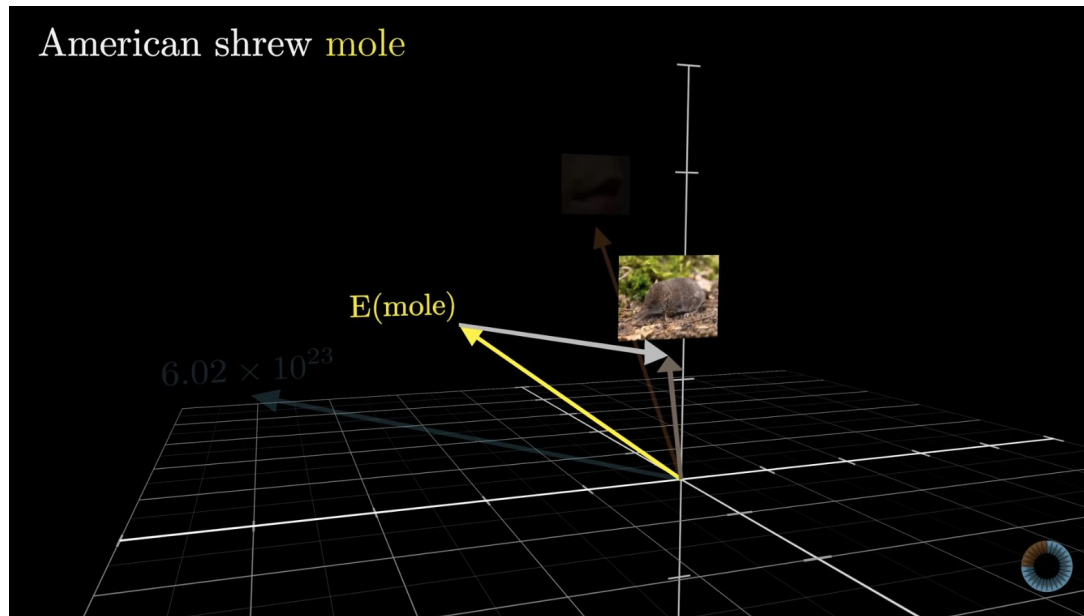
The attention pattern (this table) gives us a score: how relevant every token is updating every other token

(Then we softmax columns so that they're between 0 and 1 and can be used as a weight)

Questions so far?

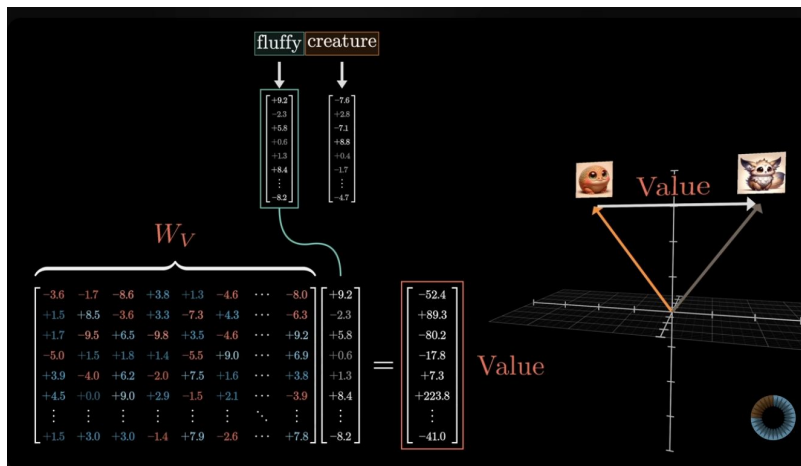
Next: remember our goal

- We want each token's starting embedding to soak up all the context
- This means we want to figure out how much to move the initial embedding into the correct, context-rich embedding



Value vector: what we add to change a token's embedding

- “If this word is relevant to adjusting the meaning of something else, what exactly should be added to the embedding of that something else in order to reflect this?”
- We get some value vector—what is the value vector modifying in this example?



Calculating value vectors: embeddings and value matrix

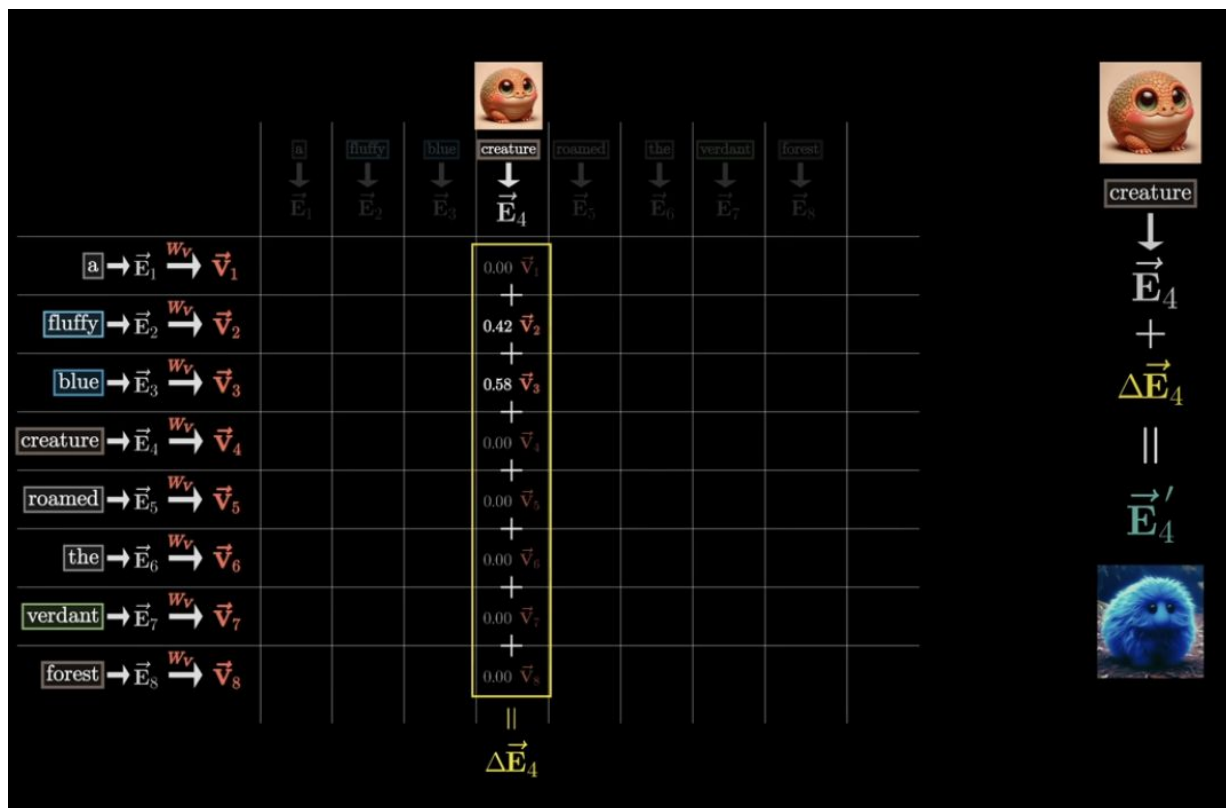
Value matrix W_V		a ↓ $\vec{E}_1$	fluffy ↓ $\vec{E}_2$	blue ↓ $\vec{E}_3$	creature ↓ $\vec{E}_4$	roamed ↓ $\vec{E}_5$	the ↓ $\vec{E}_6$	verdant ↓ $\vec{E}_7$	forest ↓ $\vec{E}_8$	
a	$\vec{E}_1 \xrightarrow{W_V} \vec{V}_1$	●	●	●	●	●	●	●	●	
fluffy	$\vec{E}_2 \xrightarrow{W_V} \vec{V}_2$		●	●	●	●	●	●	●	
blue	$\vec{E}_3 \xrightarrow{W_V} \vec{V}_3$			●	●	●	●	●	●	
creature	$\vec{E}_4 \xrightarrow{W_V} \vec{V}_4$				●	●	●	●	●	
roamed	$\vec{E}_5 \xrightarrow{W_V} \vec{V}_5$					●	●	●	●	
the	$\vec{E}_6 \xrightarrow{W_V} \vec{V}_6$						●	●	●	
verdant	$\vec{E}_7 \xrightarrow{W_V} \vec{V}_7$							●	●	●
forest	$\vec{E}_8 \xrightarrow{W_V} \vec{V}_8$								●	●

What do we do with the value vector?

Value matrix W_V		a ↓ $\vec{E}_1$	fluffy ↓ $\vec{E}_2$	blue ↓ $\vec{E}_3$	creature ↓ $\vec{E}_4$
a → $\vec{E}_1 \xrightarrow{W_V} \vec{v}_1$					0.00 $\vec{v}_1$
fluffy → $\vec{E}_2 \xrightarrow{W_V} \vec{v}_2$					0.42 $\vec{v}_2$
blue → $\vec{E}_3 \xrightarrow{W_V} \vec{v}_3$					0.58 $\vec{v}_3$

- 1) For a given column in the attention pattern...
 - Why column? Each column came from a query—thing that is looking for updates from the keys
- 2) Look at the largest numbers
 - This tells us which embeddings came from keys relevant to that query
- 3) These are the ones that update the value vectors for a given embedding

Updated embedding



Congratulations!

You now understand the core equation in the “Attention is All you Need” Paper.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{K^T Q}{\sqrt{d_k}}\right) V$$

	Q_1	Q_2	Q_3	Q_4	Q_5	$\dots$	Q_n	
K_1	$\frac{Q_1 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_1}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_1}{\sqrt{d_k}}$	
K_2	$\frac{Q_1 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_2}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_2}{\sqrt{d_k}}$	
K_3	$\frac{Q_1 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_3}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_3}{\sqrt{d_k}}$	
K_4	$\frac{Q_1 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_4}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_4}{\sqrt{d_k}}$	
K_5	$\frac{Q_1 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_5}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_5}{\sqrt{d_k}}$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	



Remaining time: Practice oral assessments with each other

- Assignments page on course website.